CSE STUDY HUB

DATA COMMUNICATION & NETWORKING UNIT 3

Network Devices, Switching & IP Addressing

SOURCE-ALIGNED CONCISE NOTES

Prepared using the uploaded handwritten class notes for this unit. The explanations retain the concepts and examples visible in those notes, with wording cleaned up for revision.

Topics covered in this unit

- Network devices and their roles
- Switching loops and Spanning Tree Protocol
- IPv4 addressing and subnetting
- Subnetting worked method and revision

How to use: Learn the definition, draw the diagram/flow where applicable, practise the numerical method, and finish with the revision questions. If a detail is not present in the uploaded pages, it is not added as an asserted class-note topic.

Exam answer structure

For a 5-mark answer: definition → labelled diagram/formula → working or components → example/application → advantage, limitation or comparison.

UNIT 3 · Network devices and their roles

A repeater regenerates or strengthens a weakened signal so it can travel farther. It does not decide where a frame should go; it repeats the signal.

A hub is a multiport repeater. It sends incoming data out through its other ports, so devices share the medium and unnecessary traffic is repeated.

A bridge connects LAN segments and filters/forwards frames using MAC addresses. A switch is a multiport bridge that forwards frames toward the appropriate port using its MAC address table.

A router works at the Network Layer and forwards packets between different networks using IP addresses and routing information. A switch typically connects devices within a LAN; a router connects networks.

UNIT 3 · Switching loops and Spanning Tree Protocol

Redundant links improve availability but can create Layer 2 loops. Loops may cause broadcast storms, duplicate frames and unstable MAC address learning.

Spanning Tree Protocol (STP) prevents switching loops by logically blocking selected redundant paths while keeping a loop-free active topology.

STP elects a root bridge and determines the best path toward it. Some switch ports forward traffic while selected redundant ports are placed in a blocking state. If the active path fails, the topology can reconverge.

Draw a triangle of three switches with redundant links; mark one root bridge and show one redundant link blocked to illustrate a loop-free tree.

UNIT 3 · IPv4 addressing and subnetting

An IPv4 address is 32 bits, normally written as four decimal octets (for example, 192.168.1.10). The address contains a network portion and a host portion.

A subnet mask or CIDR prefix identifies the network bits. For example, /24 means 24 network bits and 8 host bits. The number of addresses in a block is 2^(32-prefix).

For a conventional subnet, the network address has all host bits set to 0 and the broadcast address has all host bits set to 1. Usable host addresses lie between them (subject to special-purpose subnet rules).

Private IPv4 ranges commonly used inside local networks are 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16. Public addresses are globally routable subject to allocation and policy.

UNIT 3 · Subnetting worked method and revision

Example: 192.168.1.0/24 split into four equal subnets needs 2 borrowed host bits because 2^2 = 4. The new prefix is /26; each block has 64 addresses, conventionally 62 usable host addresses.

The /26 blocks are 192.168.1.0–63, .64–127, .128–191 and .192–255. Their network addresses are .0, .64, .128 and .192; broadcast addresses are .63, .127, .191 and .255.

Subnetting method: identify the required number of subnets/hosts; borrow bits; calculate the new prefix; find block size; list network, usable host range and broadcast address.

Revision questions: (1) Differentiate hub, switch and router. (2) Explain STP and why it is needed. (3) Solve a /24-to-/26 subnetting problem. (4) Differentiate MAC and IP addresses. (5) Explain private IPv4 ranges.